

CS101: Introduction to Computer Science

Course Type: DSC

Semester: I

Credits (L-T-P): 3-0-2

Academic Year: 2024-25

Course Overview

This course provides a foundation in computer science concepts.

Course Objectives

1. Understand basic programming concepts
2. Learn algorithm design
3. Master Python syntax

Course Learning Outcomes

CO	Description	Bloom's Level
CO1	Write basic Python programs	Apply
CO2	Analyze algorithms for efficiency	Analyze
CO3	Debug code effectively	Apply

Course Content

Unit 1: Introduction to Programming

10 Hours

- Variables and Data Types
- Control Structures

Functions

Defining and calling functions

- Parameters
- Return values
- Scope

Key Concepts: Modular programming, Abstraction

- Lists
- Dictionaries
- Sets

Teaching-Learning Methodology

Teaching Methods:

- Lectures
- Live Coding

Learning Activities:

- Assignments
- Projects

Assessment Pattern

Component	Weightage (%)
Continuous Assessment	40
Semester End Exam	60

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4
CO1	3	2	-	-
CO2	2	-	3	-
CO3	3	-	-	2

Correlation Levels: 1 = Low, 2 = Medium, 3 = High

References

Textbooks:

- Python Crash Course

Reference Books:

- Learning Python